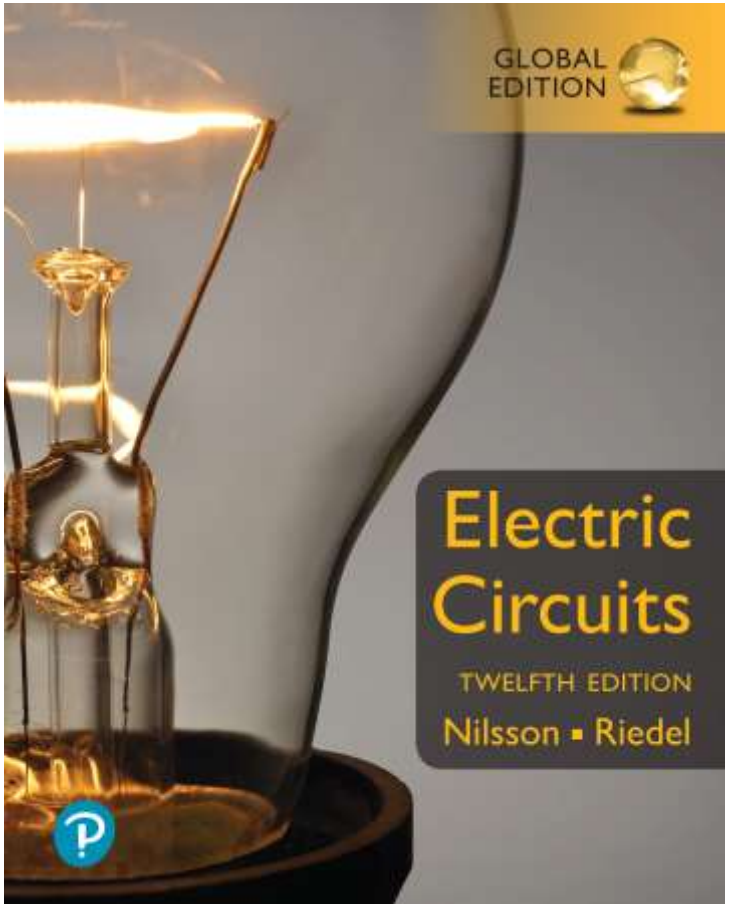
Electric Circuits

Twelfth Edition, Global Edition



Chapter 7 – Part 1

Response of First-Order *RL* and *RC* Circuits

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Chapter 7 – Response of First-Order *RL*/*RC* Circuits

Objectives:

7.1 Natural response of RL/RC circuits

7.2 Step response of RL/RC circuits

7.3 Sequential switching in first-order circuits

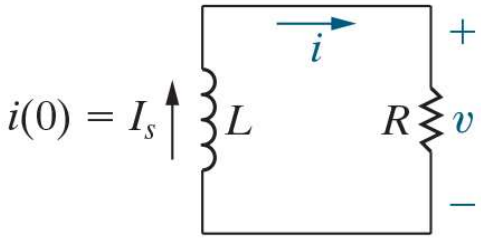
7.4 Unbounded response

7.5 Integrating amplifier

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Chapter 7 – *RL* Natural Response
(1 of 7)



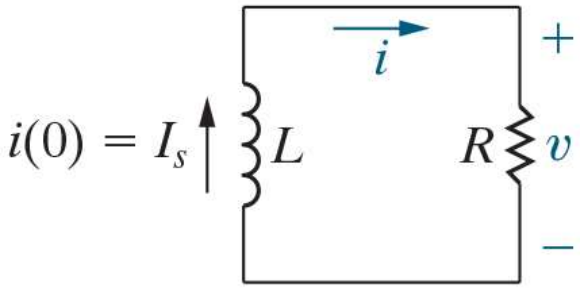
The problem: Given an initial current *i(0)* through *L* at *t = 0*, find *i(t)* for $t \geq 0$.

Note: this is the "natural" response of this circuit because for $t \geq 0$ there is no independent source in the circuit – the response is due only to the "nature" of the components and their interconnection.

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Chapter 7 – *RL* Natural Response
(2 of 7)



KVL: $L\dfrac{di(t)}{dt} + Ri(t) = 0$

- This equation is a
  - First-order (the highest derivative is a first derivative)
  - Homogeneous (the right-hand side is 0)
  - Ordinary differential equation
  - With constant coefficients

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Chapter 7 – *RL* Natural Response
(3 of 7)

The solution to the differential equation, repeated below, must be a function whose first-derivative has the same form as the function. Which one of these functions exhibits this behavior?

$$\text{KVL:} \qquad L\frac{di(t)}{dt} + Ri(t) = 0$$

A. $K \sin \omega t$

B. Ke^{-at}

C. Kt^2

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Chapter 7 – *RL* Natural Response
(3 of 7)

The solution to the differential equation, repeated below, must be a function whose first-derivative has the same form as the function. Which one of these functions exhibits this behavior?

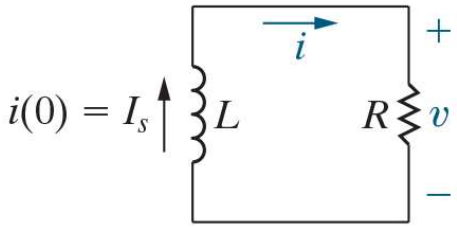
$$\text{KVL:} \qquad L\frac{di(t)}{dt} + Ri(t) = 0$$

- ✗ A. $K \sin \omega t$
- ✓ B. Ke^{-at}
- ✗ C. Kt^2

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Chapter 7 – *RL* Natural Response
(4 of 7)



KVL: $L\dfrac{di(t)}{dt} + Ri(t) = 0$

Let $i(t) = e^{-at}$ and $\dfrac{di(t)}{dt} = -ae^{-at}$

Substituting, $L(-ae^{-at}) + \mathrm{R}e^{-at} = 0$

Rearranging, $(R - aL)e^{-at} = 0$ so $R - aL = 0$

Solving, $a = R/L$

But $i(0) = I_s$ so $i(t) = I_s e^{-(R/L)t}, \quad t \geq 0$

Check: $L(-R/L)I_s e^{-(R/L)t} + RI_s e^{-(R/L)t}$

$= -RI_s e^{-(R/L)t} + RI_s e^{-(R/L)t} = 0$

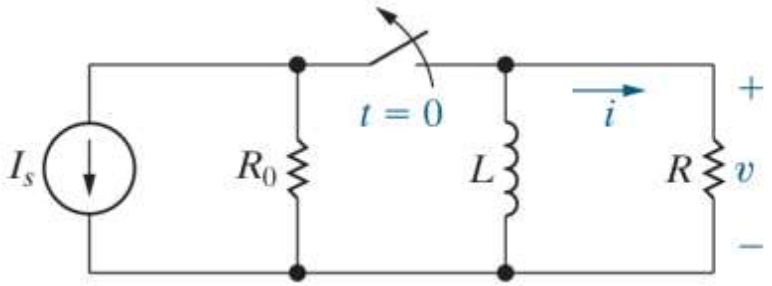
$i(0) = I_s e^{-(R/L)(0)} = I_s$

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Chapter 7 – *RL* Natural Response
(5 of 7)

Here is one circuit that can be used to establish the initial current through the inductor. For $t < 0$, the inductor can be replaced by



A. An open circuit

B. A short circuit

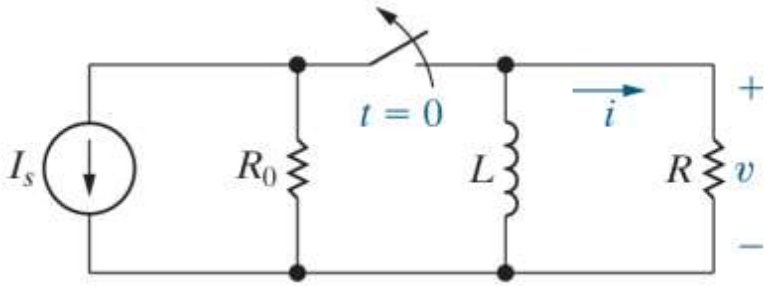
C. A resistor

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Chapter 7 – *RL* Natural Response
(5 of 7)

Here is one circuit that can be used to establish the initial current through the inductor. For $t < 0$, the inductor can be replaced by



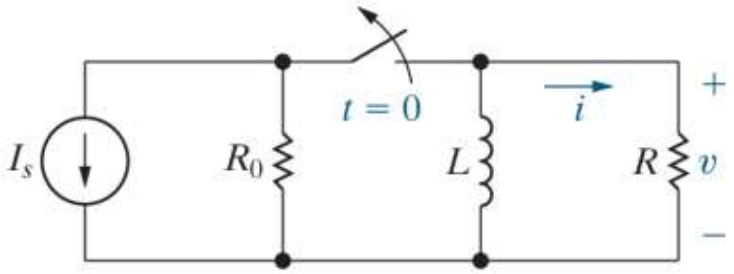
- ✗ A. An open circuit
- ✓ B. A short circuit
- ✗ C. A resistor

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Chapter 7 – *RL* Natural Response
(6 of 7)

To evaluate the current in the inductor for $t < 0$, close the switch and replace the inductor with a short circuit. The current from bottom to top in the short circuit is



A. $-I_S$

B. $RI_S\,/\,(R + R_0)$

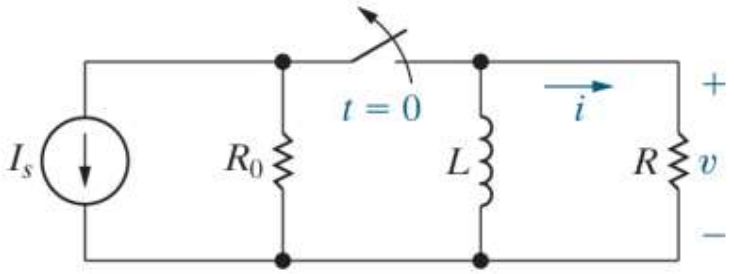
C. I_S

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Chapter 7 – *RL* Natural Response
(6 of 7)

To evaluate the current in the inductor for $t < 0$, close the switch and replace the inductor with a short circuit. The current from bottom to top in the short circuit is



✗ A. $-I_S$

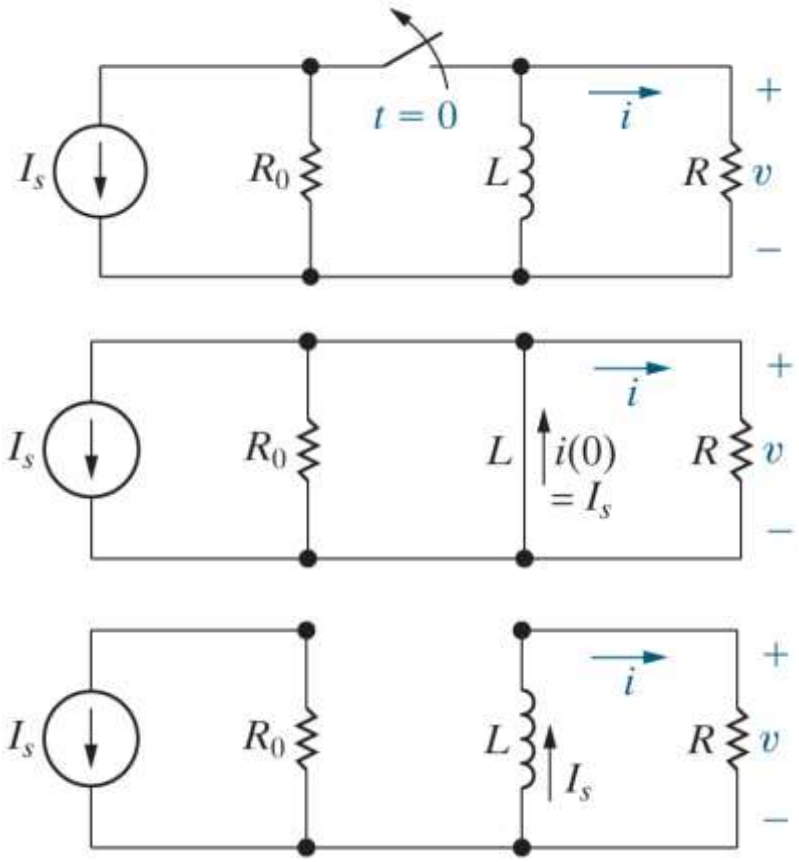
✗ B. $RI_S\,/\,(R+R_0)$

✓ C. I_S

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Chapter 7 – *RL* Natural Response
(7 of 7)

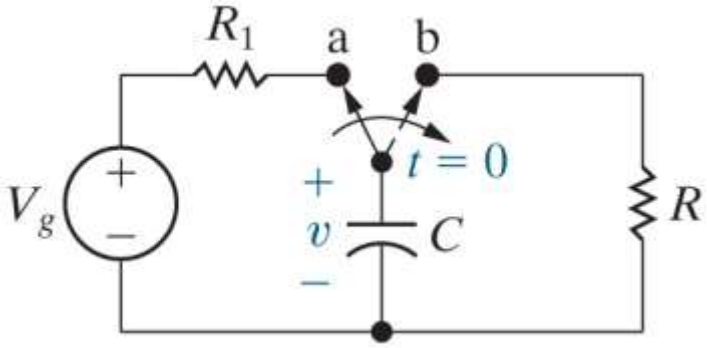


$$i(t) = I_s e^{-(R/L)t}, \qquad t \geq 0$$

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Chapter 7 – *RC* Natural Response
(1 of 5)



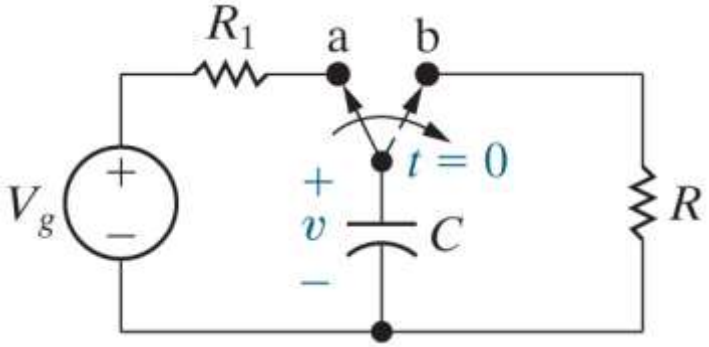
The problem: Find *v(t)* for $t \geq 0$. Note that the circuit to the left of the capacitor establishes an initial voltage drop *v(0)* across *C* at $t = 0$.

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Chapter 7 – *RC* Natural Response
(2 of 5)

Here is the circuit for the *RC* natural response problem. For $t < 0$, the capacitor can be replaced by



A. An open circuit

B. A short circuit

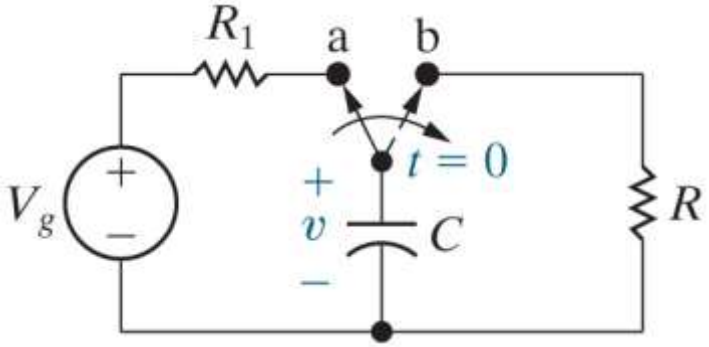
C. A constant current

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Chapter 7 – *RC* Natural Response
(2 of 5)

Here is the circuit for the *RC* natural response problem. For $t < 0$, the capacitor can be replaced by



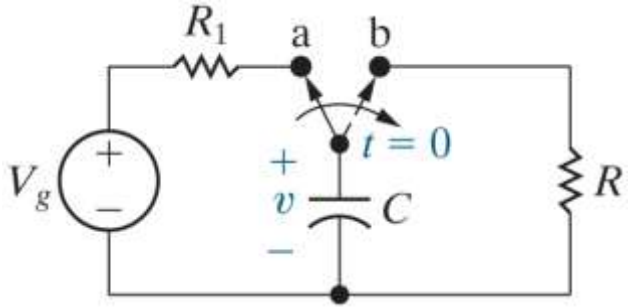
- ✔ A. An open circuit
- ✘ B. A short circuit
- ✘ C. A constant current

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Chapter 7 – *RC* Natural Response
(3 of 5)

To evaluate the voltage drop across the capacitor for $t < 0$, put the switch in the "a" position and replace the capacitor with an open circuit. The voltage drop across the open circuit, positive at the top, is



A. V_g

B. $-V_g$

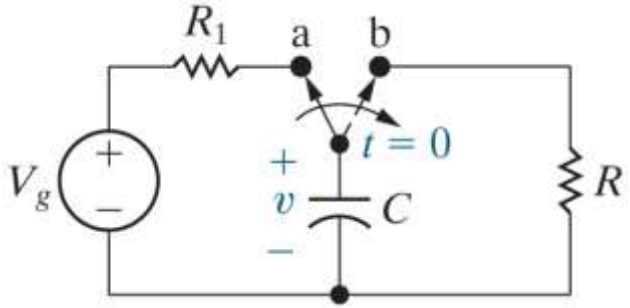
C. $RV_g / (R + R_1)$

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Chapter 7 – *RC* Natural Response
(3 of 5)

To evaluate the voltage drop across the capacitor for $t < 0$, put the switch in the "a" position and replace the capacitor with an open circuit. The voltage drop across the open circuit, positive at the top, is



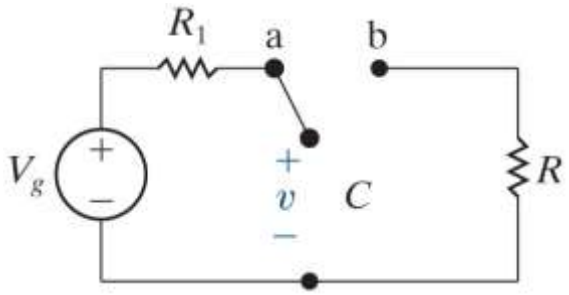
- ✔ A. V_g
- ✘ B. $-V_g$
- ✘ C. $RV_g / (R + R_1)$

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Chapter 7 – *RC* Natural Response
(4 of 5)

$t < 0$:



There is an open circuit here, so the current in this circuit is 0, and 0, and the voltage drop across R_1 is 0. Thus,

$$v = V_g, \qquad t < 0$$

Remember the continuity requirement for capacitors – the voltage drop across the capacitor must be continuous everywhere, so that the current through the capacitor remains finite. Thus,

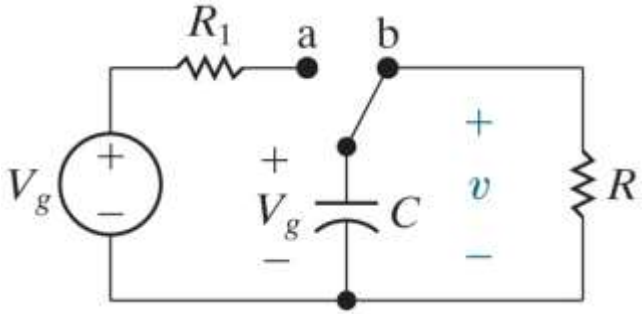
$$v(0) = V_g$$

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Chapter 7 – *RC* Natural Response
(5 of 5)

$t \geq 0$:



KCL: $C\dfrac{dv(t)}{dt} + \dfrac{v(t)}{R} = 0$

Let $v(t) = e^{-at}$ and $\dfrac{dv(t)}{dt} = -ae^{-at}$

Substituting, $C(-ae^{-at}) + e^{-at}/R = 0$

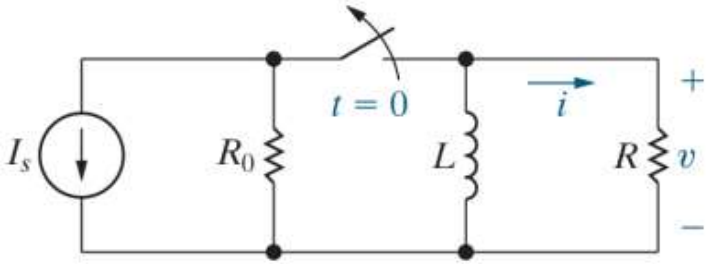
Rearranging, $(1/R - aC)e^{-at} = 0$ so $1/R - aC = 0$

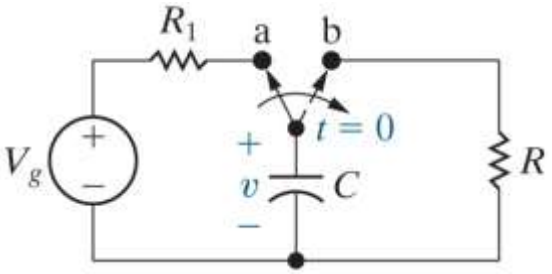
Solving, $a = 1/RC$

But $v(0) = V_g$ so $v(t) = V_g e^{-(1/RC)t}$, $t \geq 0$

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KVL: $L\dfrac{di(t)}{dt} + Ri(t) = 0$

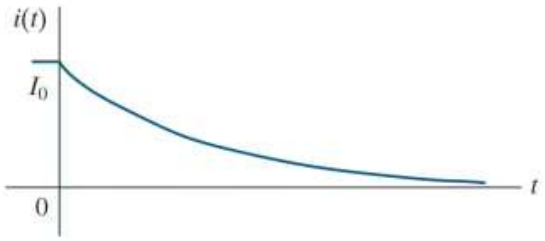
KCL: $C\dfrac{dv(t)}{dt} + \dfrac{v(t)}{R} = 0$

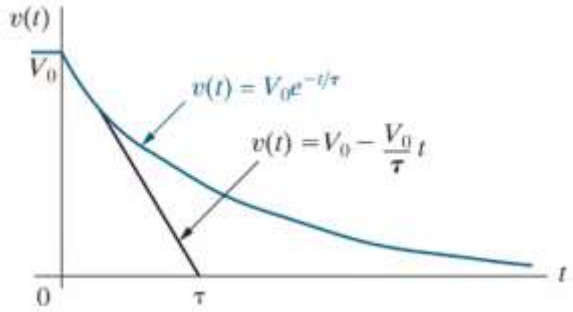
$i(0) = I_s$

$v(0) = V_g$

$i(t) = I_s e^{-(R/L)t}, \qquad t \geq 0$

$v(t) = V_g e^{-(1/RC)t}, \qquad t \geq 0$





The form of the natural response is the same:

$$ICe^{-t/\tau}$$

IC is the initial condition and τ is the **time constant**, a measure of how quickly or slowly the exponential decays.

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Chapter 7 – RL/RC Natural Response (2 of 3)

The form of the *RL*/*RC* natural response is

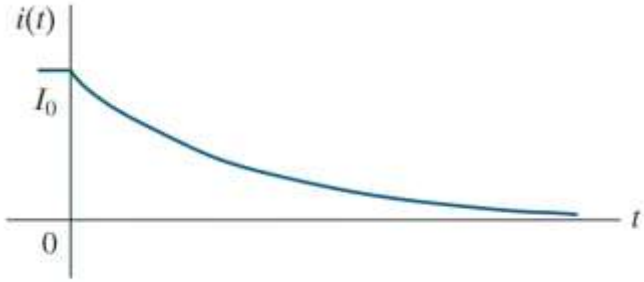
$$ICe^{-t/\tau}$$

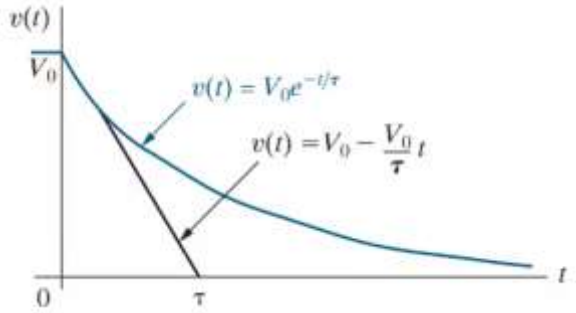
The units for the time constant, τ, must be

A. s (seconds)

B. s^{-1} (inverse seconds)

C. τ has no units





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Chapter 7 – RL/RC Natural Response
(2 of 3)

The form of the *RL*/*RC* natural response is

$$ICe^{-t/\tau}$$

The units for the time constant, τ, must be

- ✔ A. s (seconds)
- ✘ B. s^{-1} (inverse seconds)
- ✘ C. τ has no units

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Chapter 7 – RL/RC Natural Response (3 of 3)

The *RL* natural response is

$$i(t) = I_s e^{-(R/L)t}, \quad t \geq 0$$

Therefore, the time constant, τ, for the *RL* circuit is

A. R/L

B. *(L/R)t*

C. *(L/R)*

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Chapter 7 – RL/RC Natural Response
(3 of 3)

The *RL* natural response is

$$i(t) = I_s e^{-(R/L)t}, \quad t \geq 0$$

Therefore, the time constant, τ, for the *RL* circuit is

- ✗ A. R/L
- ✗ B. *(L/R)t*
- ✓ C. *(L/R)*

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